1.1 Numerical Calculations

1.1.1 Arithmetic

You can do arithmetic with *Mathematica* just as you would on an electronic calculator.

This is the sum of two numbers.	`In[1]:= 2.3 + 5.63` `Out[1]= 7.93`
Here the / stands for division, and the ^ stands for power.	`In[2]:= 2.4 / 8.9 ^ 2` `Out[2]= 0.0302992`
Spaces denote multiplication in *Mathematica*. You can use a * for multiplication if you want to.	`In[3]:= 2 3 4` `Out[3]= 24`
You can type arithmetic expressions with parentheses.	`In[4]:= (3 + 4) ^ 2 - 2 (3 + 1)` `Out[4]= 41`
Spaces are not needed, though they often make your input easier to read.	`In[5]:= (3+4)^2-2(3+1)` `Out[5]= 41`

x^*y*	power
-*x*	minus
x/*y*	divide
x *y* *z* or *x****y****z*	multiply
x+*y*+*z*	add

Arithmetic operations in *Mathematica*.

Arithmetic operations in *Mathematica* are grouped according to the standard mathematical conventions. As usual, `2 ^ 3 + 4`, for example, means `(2 ^ 3) + 4`, and not `2 ^ (3 + 4)`. You can always control grouping by explicitly using parentheses.

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This result is given in scientific notation. Remember that spaces stand for multiplication.

```
In[6]:= 2.4 ^ 45

                   17
Out[6]= 1.28678 10
```

You can enter numbers in scientific notation like this.

```
In[7]:= 2.3 10^-70

               -70
Out[7]= 2.3 10
```